

Practice Sheet

Q1. Suppose, you need to configure GPIO Port B's pin 3 as output port. Write the registers that you need and the operation you need to execute for interfacing.

Sol.

We need 2 registers. RCC register for activating the GPIO port & MODER register to configure the pin as output port.

Operation:

RCC -> AHB1ENR |= 1<<1 ; [Since port B enable is at 1 no. index and we need 1 bit left shift operation]

GPIOB -> MODER |= 1<<6 ; [Since Pin 3 starts from 6 no. index [to configure 1 pin in MODER register we need 2 slots] and output mode = 1 [or 01 in binary]]

Q2. Suppose, you need to configure GPIO Port A's pin 5 as input port. Write the registers that you need and the operation you need to execute for interfacing.

Sol.

We need 2 registers. RCC register for activating the GPIO port & MODER register to configure the pin as output port.

Operation:

RCC -> AHB1ENR |= 1<<0 ; [Since port A enable is at 0 no. index and we need 0 bit left shift operation/ no left shift operation]

GPIOA -> MODER |= 0<<10 ; [Since Pin 5 starts from 10 no. index [to configure 1 pin in MODER register we need 2 slots] and input mode = 0 [or 00 in binary]]

Q3. Suppose, you need to configure GPIO Port D's pin 3 as output port. You need to send the data at "Fast Speed" and your output type should follow open drain/output should be shown in low frequency . Write the registers that you need and the operation you need to execute for interfacing.

Sol.

We need 4 registers.

1. **RCC register** for activating the GPIO port [configure GPIO Port D],
2. **MODER register** [pin 3 as output port] to configure the pin as output port
3. **OSPEEDER register** to configure speed ["Fast Speed"]
4. **OTYPER register** to configure output type [Type]

Operation:

RCC -> AHB1ENR |= 1<<3 ; [Since port D enable is at 3 no. index and we need 3 bit left shift operation]

GPIOD -> MODER |= 1<<6 ; [01/1 = Output mode, Position 6 = pin 3]

GPIOD -> OSPEEDER |= 2<<6; [10/2 = Fast speed, Position 6 = pin 3]

GPIOD -> OTyPER |= 1<<3; [1 = Opendrain, Position = pin 3]

Q4. RCC -> AHB1ENR |= (1<<2) ; // To enable/ activate GPIOx

RCC -> AHB1ENR |= (1<<3); // To enable/ activate GPIOy

GPIOx -> MODER |= 0<<0; // Setting up the mode of pins

GPIOy -> MODER |= 1<<2; // Setting up the mode of pins

i) Identify which GPIO ports are we enabling for communication.

ii) Identify which pin are we enabling of GPIOx? State the mode of the pin as well.

iii) Identify which pin are we enabling of GPIOy? State the mode of the pin as well.

Sol.**RCC Register:**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	OTGHS ULPIEN	OTGHS EN	Res.	Res.	Res.	Res.	Res.	Res.	DMA2 EN	DMA1 EN	Res.	Res.	BKP SRAMEN	Res.	Res.
	rw	rw							rw	rw			rw		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	CRC EN	Res.	Res.	Res.	Res.	GPIOH EN	GPIOG EN	GPIOF EN	GPIOE EN	GPIOD EN	GPIOC EN	GPIOB EN	GPIOA EN
			rw					rw	rw	rw	rw	rw	rw	rw	rw

MODER Register:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MODER15[1:0]		MODER14[1:0]		MODER13[1:0]		MODER12[1:0]		MODER11[1:0]		MODER10[1:0]		MODER9[1:0]		MODER8[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MODER7[1:0]		MODER6[1:0]		MODER5[1:0]		MODER4[1:0]		MODER3[1:0]		MODER2[1:0]		MODER1[1:0]		MODER0[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

i) GPIOX = GPIOC and GPIOY = GPIOD

ii) GPIOX = GPIOC, we are enabling the PIN0 of port C and the mode of PIN0 is configured as input pin.

iii) GPIOY = GPIOD, we are enabling the PIN1 of port D and the mode of PIN1 is configured as output pin.

Q5. RCC -> AHB1ENR |= (1<<4) ;

RCC -> AHB1ENR &= (1<<3);

RCC -> AHB1ENR |= (1<<4) ;

After executing the commands what will be the which port will be enabled?

Sol.

GPIO port E.

|= : OR operation

&= : AND operation

RCC Register:

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	OTGHS ULPIEN	OTGHS EN	Res.	Res.	Res.	Res.	Res.	Res.	DMA2 EN	DMA1 EN	Res.	Res.	BKP SRAMEN	Res.	Res.
	rw	rw							rw	rw			rw		
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	CRC EN	Res.	Res.	Res.	Res.	GPIOH EN	GPIOG EN	GPIOF EN	GPIOE EN	GPIOD EN	GPIOC EN	GPIOB EN	GPIOA EN
			rw					rw	rw	rw	rw	rw	rw	rw	rw

Q6.

Activate port B and D. Suppose you want to configure pin 14 of port B as a input pin and pin 10 of port D as output pin. After necessary data transmission you want to freeze these pins temporarily. Write down the commands to lock pin 14 of port B and lock the pin 10 of port D.

Sol.

RCC -> AHBENR |= (1<<1);

RCC -> AHB1ENR |= (1<<3);

GPIOB -> MODER |= (0/00<<28);

GPIOD -> MODER |= (1/01<<20);

GPIOB -> LCKR |= (1<<14);

GPIOD -> LCKR |= (1<<10);

Q7.

Two devices - X and Y are connected via port C and E. Suppose you want to set device C using pin 9 and reset device E using pin 7 of port E. Now, write down the commands.

Sol.

GPIOC -> BSRR |= (1<<9);

GPIOE -> BSRR |= (1<<23);

Q8. Mention the complementary bit positions for both set and reset operation of Q7, and their values as well.

Sol.

Complementary bit position for Set operation: 25 and value is 0

Complementary bit position for Set operation: 7 and value is 0

Q9. Suppose you want to configure an I2C protocol which requires SDA and SCL pins. You will build the system using port C and use pin no. 5,10 respectively using the same port. The alternate function number required for it will be 7. Write down the commands.

Sol.

RCC-> AHB1ENR |= (1<<2);

GPIOC -> MODER |= (2/10<<10) |= (2/10<<20);

GPIOC -> AFR[0] |= (7/0111<<20);

GPIOC -> ARF[1] |= (7/0111<<8); // here, (10-8)*4 = 8